Chapter 4

Nonlinear equations

4.1 Introduction

The pressure drop in a fluid in motion is examined. For a flow in a pipe with a circular cross section of diameter D (meter), the Reynolds number, Re, is given by

$$Re = \frac{Dv}{\nu},$$

in which v (m/s) is the average flow velocity and ν (m^2/s) is the viscosity of the fluid. The flow is called *laminar* if $Re < 2100$ (low flow velocity) and *turbulent* if $Re > 3000$. For $2100 \leq Re \leq 3000$, the flow is neither laminar nor turbulent.

For turbulent flows, the pressure drop between inflow and outflow is given by

$$P_{\text{out}} - P_{\text{in}} = \frac{\rho w L v^2}{2gD},$$

in which w is a friction coefficient, ρ (kg/m^3) is the fluid density, L (m) is the length and g (m/s^2) is the acceleration of gravity. If the fluid contains particles (sand, paper fibers), then the friction coefficient w satisfies the equation

$$\frac{1}{\sqrt{w}} = \frac{\ln(Re\sqrt{w}) + 14 - \frac{5.6}{k}}{k},$$

in which k is a parameter known from experiments.

In this chapter, numerical methods will be discussed that can be used to determine w if the values of Re and k are known.

4.2 Definitions

In this chapter, various iterative methods will be considered to solve nonlinear equations of the form $f(p) = 0$. The point p is called a *zero* of the function f, or a *root* of the equation $f(x) = 0$. First, some useful definitions and concepts are introduced.

Convergence

Each numerical method generates a sequence $\{p_n\} = p_0, p_1, p_2, \ldots$ which should converge to p: $\lim_{n\to\infty} p_n = p$. Assume that the sequence indeed converges, with $p_n \neq p$ for all n. If there exist positive constants λ and α satisfying

$$\lim_{n\to\infty} \frac{|p - p_{n+1}|}{|p - p_n|^{\alpha}} = \lambda, \tag{4.1}$$